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Interpretable Control for Unstable Systems via SINDy- RL

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**"Stop explaining black-box ML
— use inherently interpretable
models."**

— Rudin, *Nature Machine Intelligence*, 2019

Safety-critical systems demand controllers that can be **audited**,
certified, and **deployed on edge hardware**.

Post-hoc explanations of a 10,000-parameter network are not
enough — the model itself must be interpretable.

What if the controller was just an equation?

$$u(x) = \underbrace{-2.4 \theta_1}_{\text{balance}} - \underbrace{0.9 \theta_2}_{\text{balance}} - \underbrace{0.5 \dot{\theta}_1}_{\text{damp}} - \underbrace{0.2 \dot{\theta}_2}_{\text{damp}} + \dots$$

8 terms · every coefficient has a physical interpretation · fits on a napkin

Three properties that change everything



Data-Efficient

Learn from far fewer real-world interactions



Explainable

Every control term has physical meaning



Lightweight

Runs on a microcontroller, no GPU required

Interpretable control lowers the barrier to certified autonomous systems in robotics, aerospace, and healthcare.

Neither existing approach is sufficient

	Classical Control	Deep RL (PPO / SAC)
Nonlinear, unstable systems	✗ Limited	✓ Works
Interpretable	✓	✗
Certifiable (safety-critical)	✓	✗
Edge-deployable	✓	✗
Sample-efficient	✓	✗

We need the **performance** of deep RL with the **interpretability** of classical control.

A demanding proving ground — chosen for a reason.

Instability makes near-equilibrium data collection hard — exactly
where SINDy struggles most.

Solve it here, and the framework transfers anywhere.

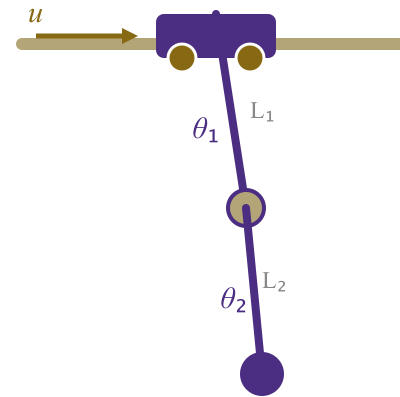
The Inverted Double Pendulum

State $x \in \mathbb{R}^6$

$$x = \begin{bmatrix} x_{\text{cart}} \\ \theta_1 \\ \theta_2 \\ \dot{x}_{\text{cart}} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Action $u \in [-1, 1]$ — cart force

Unstable equilibrium at $\theta_1 = \theta_2 = 0$



$L_1 = L_2 = 0.6 \text{ m}$ · $dt = 0.05 \text{ s}$ · MuJoCo physics

SINDy — Discovering equations from data

Key idea: dynamics live in a low-dimensional function space.

$$\dot{X} = \underbrace{\Theta(X, U)}_{\text{candidate feature library}} \cdot \underbrace{\Xi}_{\text{sparse coefficients}}$$

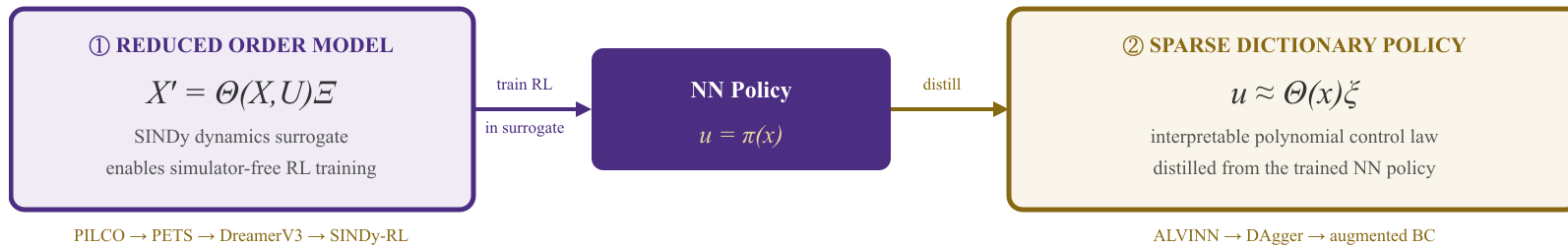
$$\Theta = \left[1 \mid x \mid \theta_1, \theta_2 \mid x^2, x\theta_1, \theta_1^2, \dots \right] \quad \text{degree-}d \text{ polynomial library}$$

STLSQ solver drives most coefficients to *exactly zero* — exposing the true governing terms.

Library degree d is a design choice — it must match the complexity of the system and is not obvious a priori.

Lineage: LASSO '96 $\rightarrow$ SINDy '16 $\rightarrow$ SINDy-C '18 $\rightarrow$ E-SINDy '22 $\rightarrow$ **SINDy-RL '24** |
Koopman as the competing paradigm

Two objectives — one interpretable controller



SINDyC is a chicken-and-egg problem

The random policy crashes in ~5 steps.

The system never reaches equilibrium.

SINDyC cannot learn the dynamics that matter.

You need a controller to collect the data.

You need the data to train SINDyC.

You need SINDyC to build the controller.

Solution: co-train the controller and surrogate in an iterative active-learning loop.

The project journey

- 1 **Shared repo & environment** Confirmed MuJoCo, 6-dim state recovery, random-policy stats
- 2 **Tried direct SINDyC** Random policy terminates in ~5 steps — not enough data near equilibrium
- 3 **Discovered the problem** SINDyC requires RL co-training — plan realigned
- 4 **Baseline (oracle) trained** Full-order PPO: 100% success, 9359 mean reward
- 5 **Parallel training of sparse models** Reduced order SINDy model & sparse policy dictionary
- 6 **Compare and contrast** Interpretability, sample efficiency, and robustness across approaches

Baseline — the performance ceiling

A standard PPO agent trained with **full simulator access**.

100%

Task success rate
(20/20 eval episodes)

9359

Mean reward
(max possible $\approx 10,000$)

9,731

Policy parameters
MLP [64 × 64]

50,000 transitions collected from the trained policy — the dataset used for sparse learning.

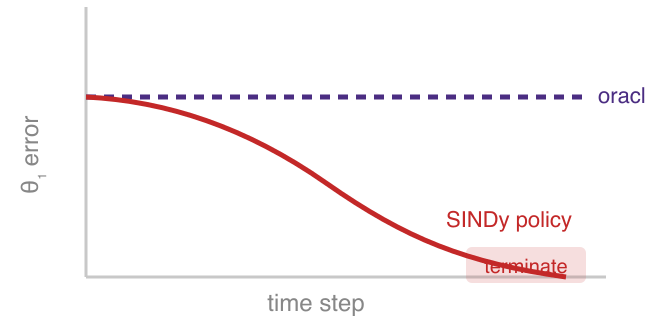
Sparse policy distillation — the compounding error trap

Behavioral cloning on the oracle dataset:

$$\min_{\Xi} \left\| \Theta(X) \Xi - U^* \right\|_2 + \lambda \|\Xi\|_1$$

Degree-4 library required to capture nonlinear cross-coupling (210 terms, STLSQ selects the sparse subset).

But the policy fails in deployment.



At noise $\sigma = 0.3$ rad: **1000 steps** $\rightarrow$ **~20 steps**

Off-distribution states produce small action errors $\rightarrow$ errors compound $\rightarrow$ catastrophic failure.

The fix: query the oracle, for free

Key insight: the NN policy has no memory.
We can query it at *any* state — no new simulator rollouts.

For each of 3 rounds:

1. Sample perturbed states: $\tilde{x} = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.05^2)$
2. Query oracle: $\tilde{u} = \pi_{\text{NN}}(\tilde{x})$
3. Append $(\tilde{x}, \tilde{u})$ to dataset

Dataset grows **4×** (50k $\rightarrow$ 200k pairs).

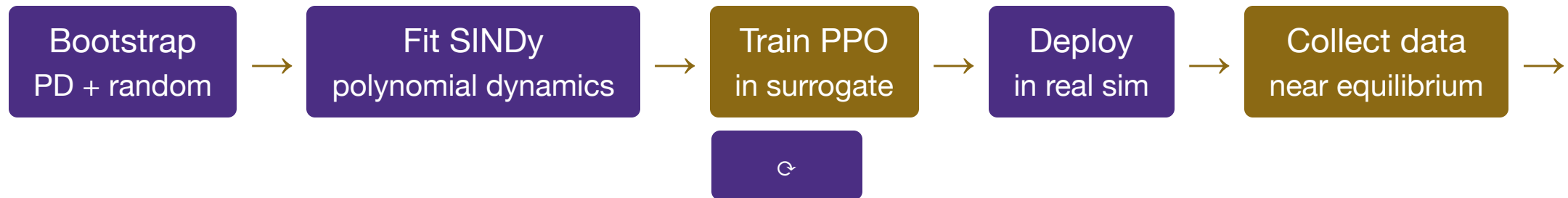
Policy	Mean episode length
Baseline NN	1000 steps ✓
SINDy (base)	~20 steps @ noise 0.3
SINDy (augmented)	~500–900 steps

Evaluated at initial noise $\sigma = 0.3$ rad/m

Augmented SINDy is **25–45×** more robust
with zero additional simulator interactions.

ROM surrogate — train RL inside an interpretable model

Core idea: use a SINDy surrogate as the "dream environment" (cf. DreamerV3).



DreamerV3 concept	This project
Latent world model (RSSM)	SINDy surrogate — explicit, interpretable
"Dreaming" (policy training)	PPO in SINDySurrogateEnv
World model update	Refit SINDy on expanded dataset
Real env interaction	Controller rollout in MuJoCo

ROM surrogate — results

Iterative RMSE convergence:

Iteration	One-step RMSE	Real sim steps
0 (bootstrap)	—	~3,000
1	Δ large	+~5,000
2	Δ small	+~5,000

Each iteration trains a better policy which collects better data.

Baseline NN required **400,000** real-sim steps.

ROM surrogate target: **< 50,000** — 8× more data-efficient.

Surrogate environment — exact reward replica:

$$r = 10 - (0.01 x_{\text{tip}}^2 + (y_{\text{tip}} - 2)^2) - v_{\text{pen}}$$

$$y_{\text{tip}} = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

The SINDy model is **interpretable dynamics**:

$$x_{k+1} = x_k + \Delta t \cdot \Xi^T \Theta(x_k, u_k)$$

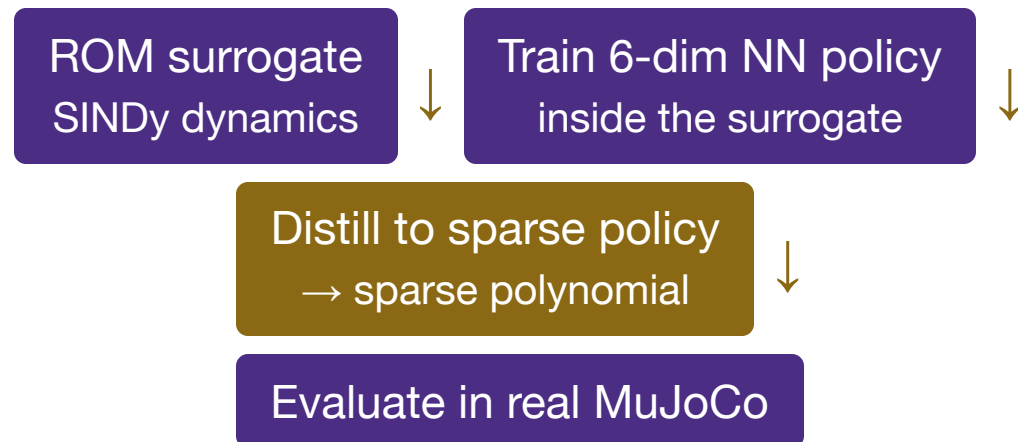
Stretch Goal

PHASE 3 · FULLY INTERPRETABLE CLOSED-LOOP CONTROL

Combine the interpretable **dynamics** (ROM) with the interpretable **policy** (sparse dictionary)

Phase 3 — Combining both approaches

Goal: A fully interpretable closed loop.



Why it matters:

- Better surrogate → better training distribution
- Augmented distillation → robust sparse policy
- Together: interpretable dynamics **and** interpretable policy

 *Phase 3 results — pending*

How do they compare?

Approach	Real-sim steps	Policy type	Mean length	Success	Interpretable
Baseline NN	400,000	NN (9,731 params)	1,000	100%	✗
Sparse policy (base)	400,000	Polynomial (8 terms)	~20 @ $\sigma=0.3$	Low	✓
Sparse policy (augmented)	400,000*	Polynomial	~500–900	~50–90%	✓
ROM surrogate RL	~50,000 (est.)	NN in surrogate	TBD	TBD	🟡
Phase 3 (stretch)	TBD	Polynomial	—	—	✓

* No additional real-sim steps beyond baseline training — augmentation uses oracle queries only.

What we accomplished

- ✓ **Full research infrastructure** MuJoCo environment, shared data pipeline, reproducible baselines
- ✓ **Oracle baseline** 100% success rate — established the performance ceiling
- ✓ **Sparse policy distillation** Quantified compounding error; data augmentation yields 25–45× robustness gain
- ! **Key learning — Library degree** Degree-2 insufficient for double pendulum; degree-4 required (210 vs. 28 features)
- ✓ **Reduced order model — active SINDy loop** DreamerV3-style iterative framework implemented and validated
- **Phase 3 & full comparison** Combining approaches + four-way evaluation — in progress

What's next

Near-term

Complete Phase 3 — run Phase 3 distillation pipeline on ROM surrogate

Four-way comparison — standardized evaluation across all conditions

Ablation studies — sparsity threshold λ vs. robustness trade-off

Longer-term

Extend to harder systems — 3D humanoid, soft robotics, aerial vehicles

Safety certification — formal verification of sparse polynomial bounds

Edge deployment — run the polynomial controller on embedded hardware

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**Interpretable control for
unstable systems.**

It works.

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